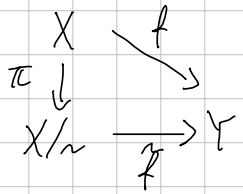
In this paper, we observe that typical manifolds.

We promise that: equivalent relation on X generated by: condition means the smallest equivalent relation on X satisfying the condition.
Rigorously, $\sim := \bigcap_{R_1 \times R_2} R_1 \times R_2$
$R_1 \times R_2 \subseteq X \times X$ is eq.rel
$R_1 \times R_2$ satisfies the condition

And, $I = [0,1]$, the subspace of the euclidean space $\mathbb{R}$.



Cylinder

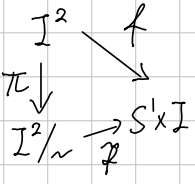
Def.

Define an equivalent relation $\sim$ on $I \times I$ generated by: $\forall y \in I,\ (0,y) \sim (1,y)$
The quotient space $I^2/\sim$ is called the Cylinder.

Prop. The cylinder $I^2/\sim$ is Homeomorphic to $S^1 \times I \subseteq \mathbb{C} \times \mathbb{R}$.

Proof. Define a map

$$f: I^2 \to S^1 \times I : (x,y) \mapsto \left(e^{2\pi i \cdot x}, y\right)$$



Then, f is continuous onto closed map, because

1. Continuous

$(x,y) \mapsto e^{2\pi i \cdot x}$ and $(x,y) \mapsto y$ are continuous, so f is also continuous.

2. Onto

Given $(e^{i\theta}, b) \in S^1 \times I$ where $\theta \in [0, 2\pi]$, take $x = \frac{\theta}{2\pi}$ and $y = b$, then clear.

3. Closed

Given closed set $C \subseteq I^2$, C is compact since I^2 is compact.
So, the continuous image $f[C]$ is also compact in $S^1 \times I$,
therefore $f[C]$ is closed, by $S^1 \times I$ is Hausdorff.

Moreover, $f(x_1,y_1) = f(x_2,y_2) \iff e^{2\pi i x_1} = e^{2\pi i x_2}$ and $y_1 = y_2$

$$\left(\iff e^{2\pi i (x_1 - x_2)} = 1\right)$$

$$\iff \left(x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0,1\}\right) \text{ and } y_1 = y_2$$

$$\iff (x_1,y_1) \sim (x_2,y_2)$$

Therefore, f induces a Homeomorphism

$$\tilde{f}: I^2/\sim \to S^1 \times I : [(x,y)] \mapsto f(x,y).$$

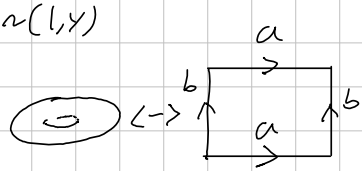
It is injective and well-defined by $(x_1,y_1) \sim (x_2,y_2) \iff f(x_1,y_1) = f(x_2,y_2)$
And surjective is given by the f is surjective.
Finally, the continuity is given by: $\tilde{f} \circ \pi = f$ is continuous, and,
given closed set $C \subseteq I^2/\sim$, $\pi^{-1}[C]$ is closed, so the ontoness of π gives
$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$ is closed, since f is closed map. ☒

Torus

Def.

Define an equivalent relation $\sim$ on $I \times I$ generated by:

$$\begin{cases} \forall x, y \in I, \ (x,0) \sim (x,1) \text{ and } (0,y) \sim (1,y) \\ (0,0) \sim (1,0) \sim (0,1) \sim (1,1) \end{cases}$$



The quotient space $I^2/\sim$ is called a Torus.

Prop. The Torus $I^2/\sim$ is Homeomorphic to $S^1 \times S^1 \subseteq \mathbb{C}$.

Proof. Define a map

$$f: I^2 \to S^2 : (x,y) \mapsto (e^{2\pi i \cdot x}, e^{2\pi i \cdot y})$$

Then, f is continuous onto closed map,

$\langle a, b \mid aba^{-1}b^{-1} = e \rangle$

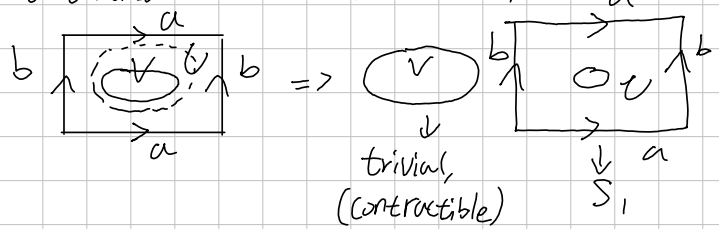
$\cong \mathbb{Z} \times \mathbb{Z}$

$I^2/\sim \cong S^1 \times S^1$ gives

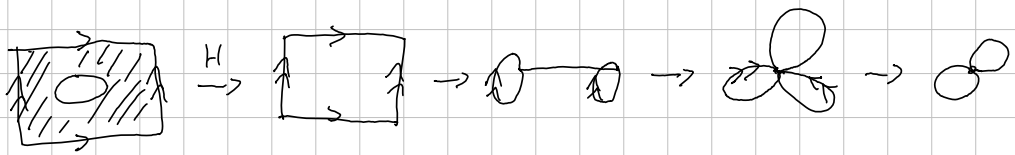
$\pi_1(I^2/\sim) \cong \pi_1(S^1 \times S^1)$

$\cong \pi_1(S^1) \times \pi_1(S^1)$

Fundamental group of the Torus,







Therefore, $\pi_1(U) * \pi_1(V) / \pi_1(U \cap V)$

$\cong \langle a \rangle * \langle b \rangle / \langle aba^{-1}b^{-1} \rangle$

$\cong \mathbb{Z} \times \mathbb{Z}$.

Projective Space.

Def. Define an equivalent relation on $\mathbb{R}^{n+1} \setminus \{0\}$ generated by:

$$x \sim y \iff \exists t \in \mathbb{R} \text{ such that } tx = y.$$

The quotient space $\mathbb{RP}^n \overset{\text{def}}{=} (\mathbb{R}^{n+1} \setminus \{0\}) / \sim$ is called a **Real Projective Space.**

[Prop. $\mathbb{RP}^1$ is Homeomorphic to the circle S^1

Proof. Define a map

$$f:\mathbb{R}^2\setminus\{(0,0)\}\to S^1;(x,y)\mapsto \frac{1}{x^2+y^2}\cdot(x^2-y^2,2xy)$$

It is derived by $r\cdot e^{i\theta}\mapsto e^{i\cdot 2\theta}$, so f is onto. Moreover, it is continuous map, being rational function, entrywise.

And, observe that

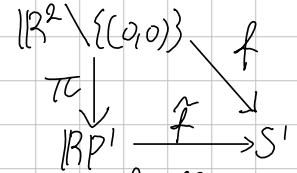
$(x_1,y_1)\sim(x_2,y_2) \iff \theta_1-\theta_2=k\pi$ for some $k\in\mathbb{Z}$ where $e^{i\theta_j}=(x_j,y_j)$ $(j=1,2)$.

$$\iff e^{i\cdot 2(\theta_1-\theta_2)}=1$$

$$\iff f(x_1,y_1)=f(x_2,y_2).$$

Therefore, the induced map

$$\tilde{f}:\mathbb{RP}^1\to S^1:[(x,y)]\to f(x,y)$$



$f=\tilde{f}\circ\pi$.

is well-defined, injective, surjective.

Moreover, for $i:S^1\to\mathbb{R}^2\setminus\{(0,0)\};(x,y)\mapsto(x,y)$,

$$\pi\circ i:S^1\to\mathbb{RP}^1$$

is continuous onto map, and since S^1 is compact, so $\mathbb{RP}^1$ is compact.

Finally, $\mathbb{RP}^1$ is compact and S^1 is Hausdorff, thus f is closed map, therefore $\tilde{f}$ is also closed map, so $\mathbb{RP}^1\cong S^1$ □

Covering space of $\mathbb{R}P^n$.

covering map.

Thm. Let $X \sim -X$ $(X \in S^2)$ be an equivalent relation.
Then, $\mathbb{R}P^2 \cong S^2/\sim$ and the natural Projection

$$\pi : S^2 \to S^2/\sim (\cong \mathbb{R}P^2) : X \mapsto [X]$$

is a Covering map.

Proof. Clearly, π is Continuous onto map. Therefore, enough to show that; given $[b] \in \mathbb{R}P^2$, there exists evenly covered open neibor Containing $[b]$.
To show this, notice that $\pi^{-1}[[[b]]] = \{b, -b\}$. Define

$V_1 = \{X \in S^2 \mid \langle X, b \rangle > \frac{1}{2}\}$, (that is, V_1 is a set of all points degree $< \frac{\pi}{3}$)

$V_2 = \{X \in S^2 \mid \langle X, -b \rangle > \frac{1}{2}\} = -V_1$. Then, $V_1, V_2 \subset S^2$ are disjoint open sets.

Now, let $U = \pi[V_1]$ $(= \pi[V_2])$.

Then, $\pi^{-1}[U] = \pi^{-1}[\pi[V_1]] = V_1 \cup (-V_1) = V_1 \cup V_2$,

therefore U is open, since π is quotient map,

and $\pi|V_i \to U$ $(i = 1, 2)$ are Homeomorphism, because;

Continuous and onto are clear, and given open set $W \subset V_i$,

$\pi^{-1}[\pi|[W]] = W \cup (-W)$, open, thus $\pi|[W]$ is open.

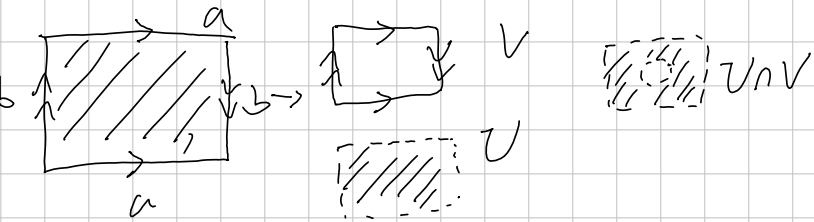
And, given $b_1, b_2 \in V_i$, $\pi|(b_1) = \pi|(b_2)$ implies that $b_1 = b_2$, because

$$\{b_1\} = [b_1] \cap V_i = [b_2] \cap V_i = \{b_2\}.$$

Consequently, π is a Covering map. □

Yeon Dong Won

Klein bottle.



$\rightarrow \pi_1(V) = \langle a \rangle * \langle b \rangle$

$\pi_1(U) = \{1\}$

$\pi_1(U \cap V) = \langle aba^{-1}b \rangle$

So, $\pi_1(K) \cong \langle a \rangle * \langle b \rangle / \langle aba^{-1}b \rangle$

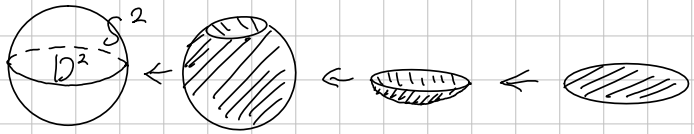
$= \langle a, b \mid aba^{-1}b = 1 \rangle$

Möbius band.

[Sphere]

Thm. $D^n/S^{n-1} \cong S^n$.

Proof. Define a map



$f: D^n \to S^n;\ x=(x_1,\cdots,x_n) \mapsto (2x_1\sqrt{1-\|x\|^2},\ 2x_2\sqrt{1-\|x\|^2},\ \cdots,\ 2x_n\sqrt{1-\|x\|^2},\ 2\|x\|^2-1)$